Multimodal LLMs (Chapter 9) — CLIP, OpenCLIP

Part 1 — What Are Multimodal LLMs?

Core Idea

A **Multimodal Model** can process **multiple types of data**, such as:

- Text
- Images
- Audio
- Video

Traditional LLM:

Text → Text

Multimodal Model:

Image + Text → Text

Image → Caption

Text → Image

Image ↔ Text Matching

Examples:

- GPT-4V
- Gemini
- CLIP

Part 2 — CLIP: Core Concept

(CLIP is one of the most important models in modern multimodal AI.)

Developed by:

- OpenAI

CLIP Training Objective

CLIP learns:

Image ↔ Text similarity

It maps:

- Image → embedding vector
- Text → embedding vector

Then:

Correct image-text pairs → HIGH similarity

Wrong pairs → LOW similarity

CLIP Architecture

Two encoders:

Image Encoder

Typically:

- ResNet OR
- Vision Transformer (ViT)

Input:

Image → Feature Vector

Text Encoder

Typically:

- Transformer

Input:

Sentence → Feature Vector

Joint Embedding Space

Both outputs mapped to same vector space:

Image embedding ≈ Text embedding

for matching pairs.

Contrastive Learning in CLIP

Suppose batch size:

N = 4

We create:

4 correct image-text pairs

12 incorrect pairs

CLIP learns:

Correct → Similar

Incorrect → Dissimilar

Loss:

Contrastive Loss

(InfoNCE)

Why CLIP Is Powerful

CLIP enables:

Zero-Shot Classification

Without training classifier:

Example:

Image → ?

Possible labels:

"cat"

"dog"

"car"

"tree"

CLIP compares:

Image embedding vs label text embeddings

Chooses:

Highest similarity

CLIP Applications

Students should know:

- Image classification
- Image search
- Image captioning

- Visual grounding
- Robotics perception
- Multimodal retrieval

Part 3 — OpenCLIP

Important real-world extension.

Developed by:

- LAION

Implementation:

- OpenCLIP

Why OpenCLIP Matters

Original CLIP:

- Closed dataset

OpenCLIP:

Uses:

LAION-400M

LAION-2B

Datasets:

- Web-scale image-text pairs

Much larger scale.

- Conceptual notes:Original CLIP:
- Closed dataset

OpenCLIP:

Uses:

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Much larger scale.

CLIP vs OpenCLIP

Feature	CLIP	OpenCLIP
Released by	OpenAI	LAION
Dataset	Private	Open
Training size	Millions	Billions
Accessibility	Limited	Open

Part 4 — Embeddings in Multimodal Models

Important connection to RAG and search.

CLIP embeddings enable:

Text → Image retrieval

Image → Text retrieval

Example:

Query:

"A red car on a snowy road"

System:

Returns matching images

Assignment 1 — CLIP Zero-Shot Classification

Difficulty: ⭐⭐
Marks: 15

Platform: Google Colab

Task

Use pretrained:

- OpenCLIP

Perform:

Zero-shot image classification

Dataset

Use:

CIFAR-10

Steps

You must:

1. Load OpenCLIP model
2. Load CIFAR-10 images

3. Create text labels:

   "a photo of a cat"

   "a photo of a dog"

   ...

4. Compute similarity
5. Predict label
6. Compute accuracy

Expected Learning

You should be able to understand:

- Multimodal embeddings
- Zero-shot learning
- Contrastive inference

Assignment 2 — Text-to-Image Retrieval

Difficulty: ⭐⭐⭐
Marks: 20

Task

Build:

Text → Image search system

Steps

1. Load image dataset
2. Compute image embeddings
3. Input text query
4. Compute similarity
5. Retrieve top-5 images

Training & Fine-Tuning LLMs (Chapters 10–12) — SFT, PEFT, LoRA, QLoRA, RLHF

Part 1 — LLM Training Pipeline

Full pipeline:

Pretraining
↓
Supervised Fine-Tuning (SFT)
↓
Alignment (RLHF)
↓
Deployment

Stage 1 — Pretraining

Objective:

Predict next token

Data:

Internet-scale text

Model learns:

- Grammar

- Knowledge
- Reasoning patterns

Stage 2 — Supervised Fine-Tuning (SFT)

Most important early step.

What Is SFT?

Train model on:

Instruction → Response

Example:

Input:
Explain gravity.

Output:
Gravity is a force...

Why SFT?

Pretrained model:

Knows language

SFT:

Learns how to follow instructions

Dataset Format

Typical format:

```
{
 "instruction": "...",
 "input": "...",
 "output": "..."
}
```

Assignment 3 — Supervised Fine-Tuning

Difficulty: ⭐⭐⭐
Marks: 20

Task

Fine-tune:

Small language model.

Recommended:

- DistilGPT-2

Dataset:

Alpaca-style instruction dataset

You must:

1. Load pretrained model
2. Tokenize dataset
3. Fine-tune
4. Evaluate responses

Part 2 — Parameter Efficient Fine-Tuning (PEFT)

Key modern idea.

Instead of:

Train all parameters

We:

Train small subset

Why PEFT?

Full fine-tuning:

Problems:

- GPU heavy
- Memory heavy
- Expensive

PEFT:

Advantages:

- Faster
- Cheaper
- Smaller checkpoints

Popular PEFT Methods

Students should know:

- Adapters
- Prefix tuning
- LoRA
- QLoRA

Part 3 — LoRA (Low-Rank Adaptation)

One of the most important techniques today.

Core Idea of LoRA

Instead of modifying:

W

We learn:

$\Delta W = A \times B$

Where:

A: small matrix
B: small matrix

So:

W_new = W + AB

Low-rank update.

Why Low Rank?

Most useful updates lie in:

Low-dimensional subspace

This saves memory.

Assignment 4 — LoRA Fine-Tuning

Difficulty: ⭐⭐⭐⭐
Marks: 25

Task

Fine-tune:

Small LLM using:

- LoRA

Recommended model:

- LLaMA 2
  (or smaller alternative)

You must:

1. Load base model
2. Add LoRA adapters
3. Fine-tune
4. Compare memory usage vs full tuning

Part 4 — QLoRA

Memory-optimized LoRA.

Uses:

Quantization + LoRA

Core Idea

Steps:

1. Quantize base model to:

4-bit

2. Train:

LoRA layers

Result:

Huge memory savings

Why QLoRA Matters

Enables:

Fine-tuning large models
on single GPU

Assignment 5 — QLoRA Experiment

Difficulty: ⭐⭐⭐⭐
Marks: 25

Task

Fine-tune using:

- QLoRA

You compare:

Method	Memory
Full FT	High
LoRA	Medium
QLoRA	Low

Part 5 — RLHF (Reinforcement Learning from Human Feedback)

One of the most important alignment methods.

Used in:

- ChatGPT

RLHF Pipeline

Step 1:

Train:

Reward Model

Step 2:

Optimize:

Policy model

Step 3:

Use:

Human preference data

Why RLHF?

Improves:

- Helpfulness
- Safety
- Alignment

Assignment 6 — Preference Optimization Simulation

Difficulty: ⭐⭐⭐
Marks: 15

Task

Simulate RLHF:

Using:

Pairwise preference dataset

You must:

1. Score responses
2. Train reward model
3. Improve output ranking

Preparations for your examinations (Theory questions):

Short Questions

1. Explain contrastive learning in CLIP.
2. Define joint embedding space.
3. What is Supervised Fine-Tuning?
4. Define Parameter-Efficient Fine-Tuning.
5. Explain LoRA architecture.

Medium Questions

1. Compare CLIP and OpenCLIP.
2. Explain LoRA mathematically.
3. Describe SFT pipeline.
4. Discuss advantages of QLoRA.

Long Questions

1. Explain CLIP architecture with training objective.
2. Describe full LLM training pipeline from pretraining to RLHF.
3. Compare Full FT, LoRA, and QLoRA.
4. Explain RLHF with reward modeling.

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